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Paper Code: TCE-813

B.Tech. (Civil Engineering)
Mid Semester Examination 2025
VIII Semester
Hydropower Engineering

Time: 1:30 Hours

MM: 50

Note:

- (i) All questions are compulsory.
- (ii) Answer any sub questions among a & b in each main question.
- (iii) Total marks for each main question are ten.

Q1. (1X10=10 Marks)

- a. A power development site has the average estimated river flows as given in the table below (flow is million m^3). Find the required capacity of the reservoir in order to produce 14500 kW continuously throughout the year at an overall efficiency of 85% under a head of 110 m.

Months	Jan	Feb	Mar	April	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Flow	68	35	77	53	15	17	35	89	92	44	54	87

(CO1)

- b. The power duration curve for a hydroelectric installation shows that 876×10^5 Kwh of electrical energy can be obtained during the year. The plant is to be designed as a peak load plant for an annual load factor 25%. Determine the capacity of station. If the plant capacity factor is 20% then find the reserve capacity of plant.

Q2. (1X10=10 Marks)

- a. Explain the necessity to predict future load demand? Describe the methods of load forecasting?
- b. Explain the various sources of energy? Give the constraints in hydropower development? (CO1)

Q3. (1X10=10 Marks)

- a. In a hydroelectric station, water is available at the rate of $175 m^3/s$ under a head of 18m the turbine run at speed of 150 r.p.m with overall efficiency of 82%. Find the number of turbine required if they have the maximum specific speed of 460.
- b. The turbine is to operate under a head of 24m at 200r.p.m the discharge is $8.5 m^3/s$. If the overall efficiency is 88%, Determine: (CO2)
 - 1. Power generated
 - 2. Specific speed of turbine
 - 3. Type of turbine

Q4. (1X10=10 Marks)

- a. Calculate the specific speed of a turbine and suggest the type of turbine required for a river having a discharge of 240L/s with an available head of 45m. Take the efficiency of turbine = 82% and speed = 450r.p.m (CO2)
- b. Explain the Diversion Canal Plant and its different layouts with a neat sketch?

Q5. (1X10=10 Marks)

- a. Discuss the status of pumped storage power plant in the world and in India? (CO2)
- b. Discuss the advantages of Pumped Storage Power Plants?

